

Practice Quiz 04B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

Set identities by logical laws; unions of families and power sets.

Complements are taken in a fixed universal set. Name a definition or law at every step of part (a).

1. (a) Prove $A \cap (B \cup B^c) = A$ by a chain of equivalences beginning with membership in the left side.

Solution For an arbitrary object x ,

$$\begin{aligned} x \in A \cap (B \cup B^c) &\iff x \in A \wedge (x \in B \vee x \in B^c) && \text{(definitions)} \\ &\iff x \in A \wedge (x \in B \vee x \notin B) && \text{(definition of complement)} \\ &\iff x \in A \wedge \text{true} && \text{(excluded middle)} \\ &\iff x \in A && \text{(identity law).} \end{aligned}$$

Therefore the two sets are equal. ■

- (b) Let $\mathcal{F} = \{\{a, b\}, \{b, c\}\}$. Compute $\bigcup \mathcal{F}$ and then $\mathcal{P}(\bigcup \mathcal{F})$.

Solution $\bigcup \mathcal{F} = \{a, b, c\}$, so

$$\begin{aligned} \mathcal{P}(\bigcup \mathcal{F}) &= \{\emptyset, \{a\}, \{b\}, \{c\}, \\ &\quad \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}. \end{aligned}$$

■